

Supplementary Material

ESM_1: Main Analysis Tables (Observed 2006-2021 Sample)

Article:

Smart City Construction, Sustainable Urban Transition, and Carbon Emissions: Nonlinear Evidence from Chinese Cities

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Descriptive statistics: observed sample, 2006-2021

	count	mean	p50	sd	min
CE	4544	4.528	4.554	1.292	0.000
SCCD	4544	0.313	0.297	0.086	0.121
Spatial coupling coordination degree squared	4544	0.106	0.088	0.068	0.015
OPEN	4529	0.193	0.076	0.345	-0.720
UR	4544	53.912	51.839	15.862	6.491
URG	4529	2.466	2.377	0.567	1.207
GI	4544	0.185	0.159	0.101	0.043
ER	4544	0.008	0.007	0.003	0.000
DEI	4544	0.056	0.042	0.077	0.000
POLY	4544	0.347	0.376	0.194	0.000
OIU	4544	2.278	2.270	0.149	1.163
GTI	4544	3.931	3.871	1.863	0.000
MCCD	4544	0.313	0.297	0.086	0.121
NTL	4544	1.380	1.385	1.153	-2.322
FDI	4513	2.252	2.285	0.273	-1.693
RD	4544	0.015	0.010	0.016	0.000
EI	4544	-0.198	-0.210	0.587	-5.026
ISL	4544	0.406	0.398	0.102	0.038
ISU	4544	0.989	0.857	0.563	0.094
IV	4260	0.317	0.301	0.087	0.121
c_SCCD	4544	-0.000	-0.016	0.086	-0.192
is_fitted	4544	0.000	0.000	0.000	0.000

Baseline regression: observed-only sample, 2006-2021

	(1) CE	(2) CE	(3) CE	(4) CE	(5) CE
SCCD	12.957*** (1.445)	11.841*** (1.421)	9.007*** (1.305)	8.819*** (1.348)	8.444*** (1.338)
Spatial coupling coordination degree squared	-11.476*** (1.749)	-10.296*** (1.761)	-8.400*** (1.593)	-8.125*** (1.642)	-8.091*** (1.665)
OPEN		0.456*** (0.108)	0.211** (0.083)	0.244*** (0.083)	0.276*** (0.082)
UR			0.029*** (0.004)	0.027*** (0.004)	0.025*** (0.004)
URG				-0.116* (0.068)	-0.101 (0.067)
GI					-1.953*** (0.546)
Constant	1.203*** (0.284)	1.314*** (0.272)	0.629** (0.267)	1.053*** (0.357)	1.497*** (0.382)
N	4544	4529	4529	4514	4514
Within R-squared	0.592	0.601	0.640	0.641	0.650

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Robustness and IV-2SLS: observed-only sample, 2006-2021

	(1) w_CE	(2) NTL	(3) CE	(4) CE
w_SCCD	9.845*** (1.337)			
w_SCCD2	-10.134*** (1.648)			
w_OPEN	0.385*** (0.117)			
w_UR	0.025*** (0.004)			
w_URG	-0.054 (0.068)			
w_GI	-1.968*** (0.548)			
SCCD		3.601*** (0.719)		17.971*** (2.816)
SCCD2		-2.765*** (0.763)		-15.757*** (2.969)
OPEN		0.264*** (0.090)	0.276*** (0.082)	0.176** (0.080)
UR		0.019*** (0.003)	0.025*** (0.004)	0.018*** (0.004)
URG		-0.055 (0.062)	-0.101 (0.067)	-0.078 (0.065)
GI		-1.668*** (0.362)	-1.953*** (0.546)	-1.587*** (0.564)
MCCD			8.444*** (1.338)	
MCCD2			-8.091*** (1.665)	
Constant	1.122***	-0.344	1.497***	-0.179

	(0.386)	(0.293)	(0.382)	(0.549)
N	4514	4514	4514	4232
Within R-squared	0.658	0.583	0.650	0.626

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

First-stage regressions: observed-only sample, 2006-2021

	(1) SCCD	(2) SCCD2
IV	0.292*** (0.078)	-0.091 (0.065)
Squared IV based on lagged SCCD	-0.027 (0.097)	0.441*** (0.091)
OPEN	0.015** (0.007)	0.008 (0.006)
UR	0.001*** (0.000)	0.001*** (0.000)
URG	0.009** (0.004)	0.011*** (0.003)
GI	-0.105*** (0.021)	-0.078*** (0.016)
Constant	0.114*** (0.022)	-0.001 (0.016)
N	4232	4232
Within R-squared	0.645	0.582

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Regional heterogeneity: observed-only sample, 2006-2021

	(1) Eastern China	(2) Central China	(3) Western China
SCCD	7.368*** (2.557)	12.447*** (2.490)	7.842*** (1.740)
SCCD2	-6.905*** (2.543)	-12.828*** (3.179)	-8.152*** (2.102)
OPEN	0.225*** (0.085)	0.416* (0.251)	0.258 (0.305)
UR	0.027*** (0.006)	0.022*** (0.006)	0.024*** (0.007)
URG	-0.069 (0.143)	-0.018 (0.121)	-0.187* (0.103)
GI	-1.098 (1.134)	-2.292*** (0.667)	-1.871** (0.885)
Constant	1.615** (0.711)	0.545 (0.700)	1.920*** (0.639)
N	1537	1585	1392
Within R-squared	0.731	0.645	0.624

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Mediation analysis: observed-only sample, 2006-2021

	(1) OIU	(2) CE with OIU	(3) GTI	(4) CE with GTI
SCCD	0.647*** (0.160)	7.561*** (1.254)	10.379*** (1.402)	5.290*** (1.115)
SCCD2	-0.407* (0.207)	-7.535*** (1.503)	-5.775*** (1.468)	-6.336*** (1.418)
OIU		1.366*** (0.316)		
GTI				0.304*** (0.027)
OPEN	0.008 (0.012)	0.265*** (0.077)	0.224*** (0.085)	0.208*** (0.071)
UR	0.004*** (0.000)	0.020*** (0.004)	0.021*** (0.004)	0.019*** (0.003)
URG	0.029*** (0.006)	-0.141** (0.066)	0.028 (0.064)	-0.109* (0.063)
GI	-0.093* (0.049)	-1.826*** (0.521)	-2.102*** (0.403)	-1.314** (0.552)
Constant	1.808*** (0.041)	-0.972 (0.697)	-0.725* (0.411)	1.718*** (0.325)
N	4514	4514	4514	4514
Within R-squared	0.622	0.658	0.826	0.686

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Moderation analysis: observed-only sample, 2006-2021

	(1) ER	(2) DEI	(3) POLY
SCCD	10.931*** (1.991)	8.869*** (1.198)	7.925*** (2.146)
SCCD2	-11.675*** (2.435)	-7.977*** (1.423)	-8.678*** (2.660)
ER	61.643 (46.315)		
SCCD # ER	-325.788 (228.112)		
SCCD2 # ER	455.053* (266.493)		
OPEN	0.278*** (0.082)	0.263*** (0.083)	0.277*** (0.082)
UR	0.025*** (0.004)	0.025*** (0.004)	0.024*** (0.004)
URG	-0.097 (0.067)	-0.090 (0.067)	-0.090 (0.067)
GI	-1.962*** (0.536)	-2.008*** (0.544)	-1.954*** (0.532)
DEI		5.874*** (2.226)	
SCCD # DEI		-25.913*** (7.936)	
SCCD2 # DEI		24.173*** (6.665)	
POLY			-1.043 (0.941)
SCCD # POLY			2.842 (4.782)
SCCD2 # POLY			0.159

			(5.485)
Constant	1.040** (0.486)	1.364*** (0.370)	1.763*** (0.487)
N	4514	4514	4514
Within R-squared	0.651	0.652	0.653
Standard errors in parentheses			
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$			